CSC2001F

Assignment 1

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FFLADA001

Foreword

- “make array” builds and runs the array implementation.
- “make bst” builds and runs the bst implementation.

Object-Oriented Design

- ***BinarySearchTree*** is a basic implementation of a binary search tree.

- ***BinarySearchTree.Node*** is a node in the binary search tree.

- ***BinarySearchTree.BSTIterator*** is an iterator allowing inorder traversal of the tree in range-based for-loops.

- The ***Console*** class was written as a wrapper around *System.out* and *System.in*.

- ***GenericsKbApp*** is the common behaviour between the bst and array implementations of the app. It allows for the specification of an enum, ***KnowledgeBaseType*** which specifies whether the app should use the array or bst implementation of the ***KnowledgeBase***. This approach was taken to ensure that the array and bst implementations maintain the same front-end interface and support the same functionality – the only difference between the two interfaces is the backend data type used to store the ***KnowledgeBase***.

- ***GenericsKbArrayApp*** is the entry-point for the array implementation of ***GenericsKbApp***.

- ***GenericsKbBSTApp*** is the entry-point for the bst implementation of ***GenericsKbApp***.

- ***KnowledgeBase*** is the abstract base-class for all knowledge base implementations.

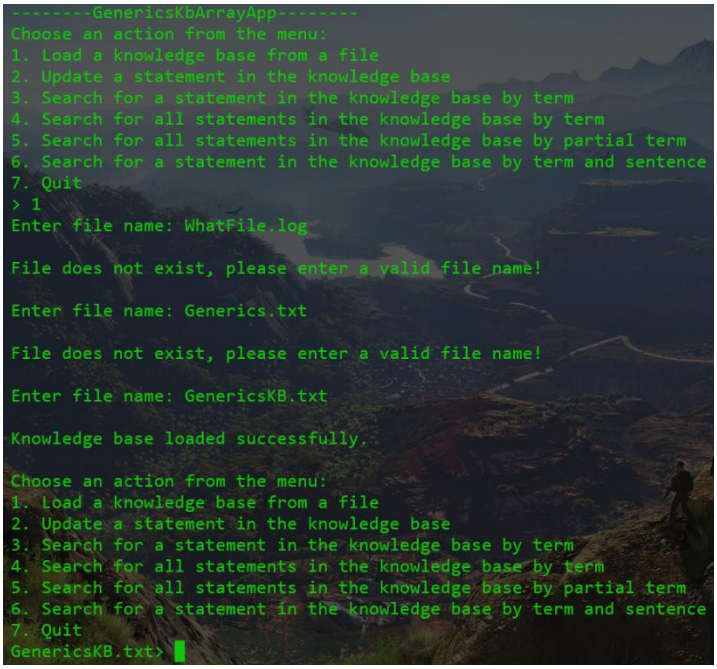
- ***KnowledgeBaseArray*** is the array implementation of the super-class ***KnowledgeBase***, which implements the knowledge base as a traditional array.

- ***KnowledgeBaseBST*** is the bst implementation of the super-class ***KnowledgeBase,*** which implements the knowledge base as a binary search tree.

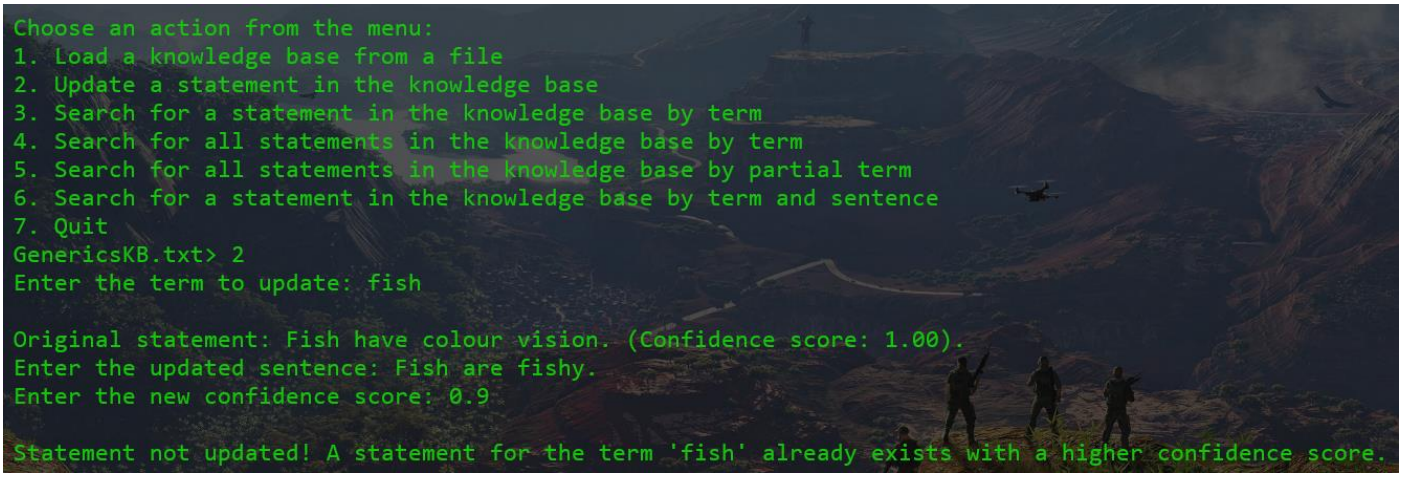
- The ***Statement*** class represents a single statement-entry in the knowledge base which contains a term, sentence and confidence score. A ***KnowledgeBase*** is a collection of ***Statements***.

Testing

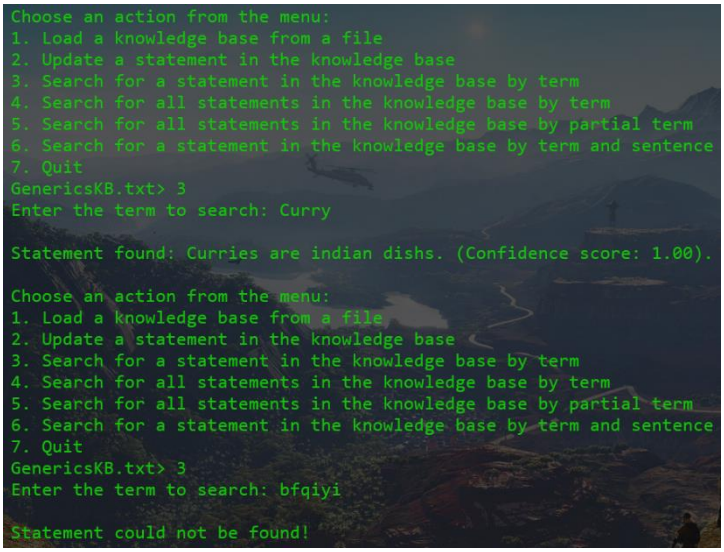
Loading knowledge base from textfile



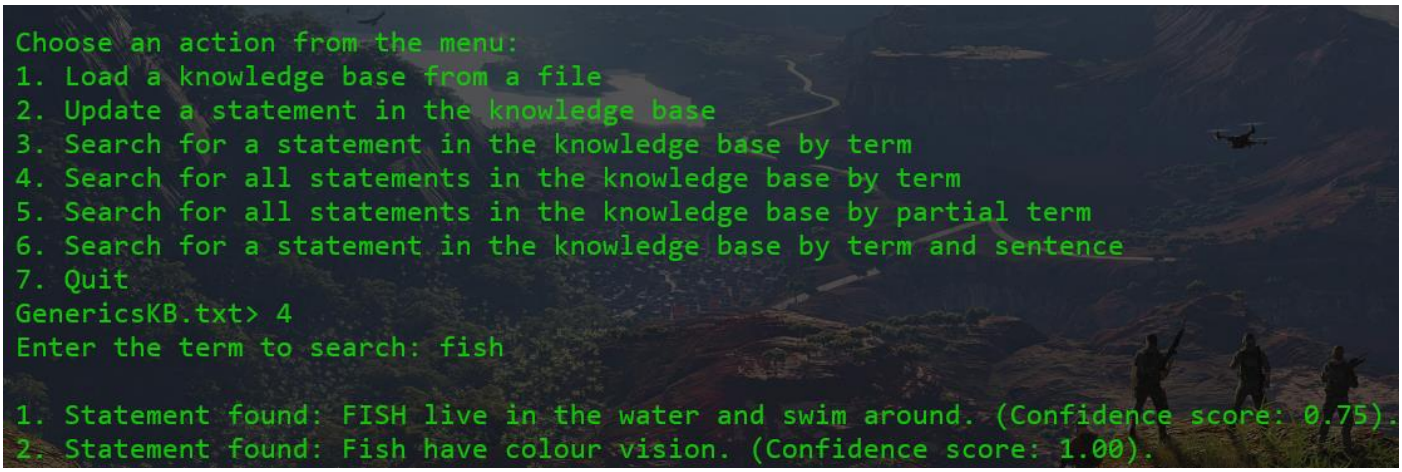
Attempting to update a statement in the knowledge base with a lower confidence score. A user-inputted confidence score exceeding 1.0 or less than 0.0 will be clamped between 0.0 and 1.0.



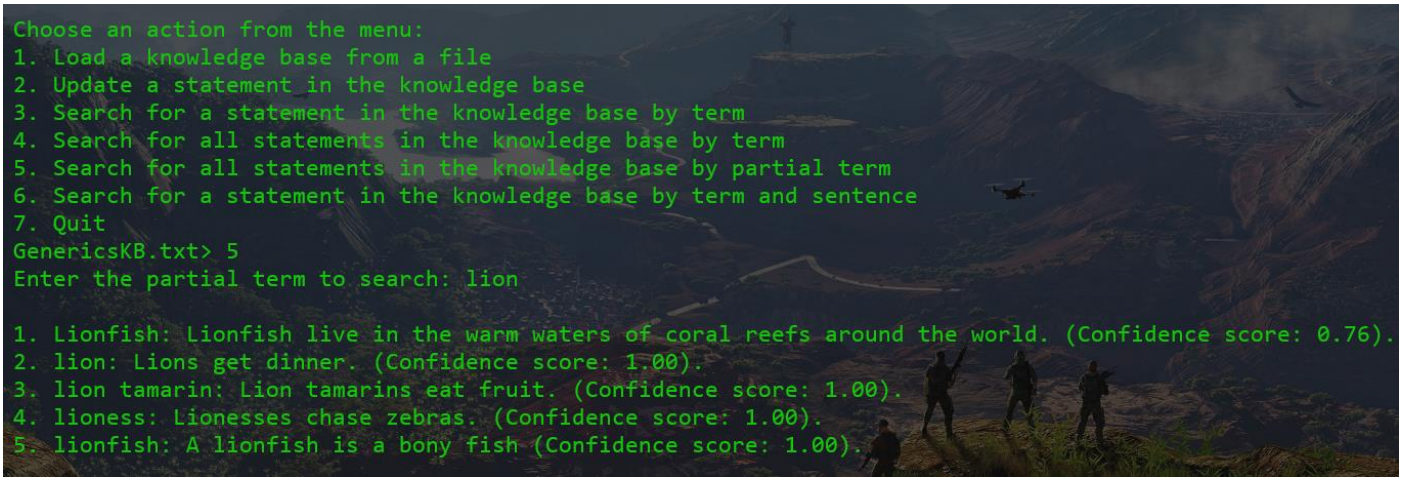
Searching for existing and non-existing statements in the knowledge base



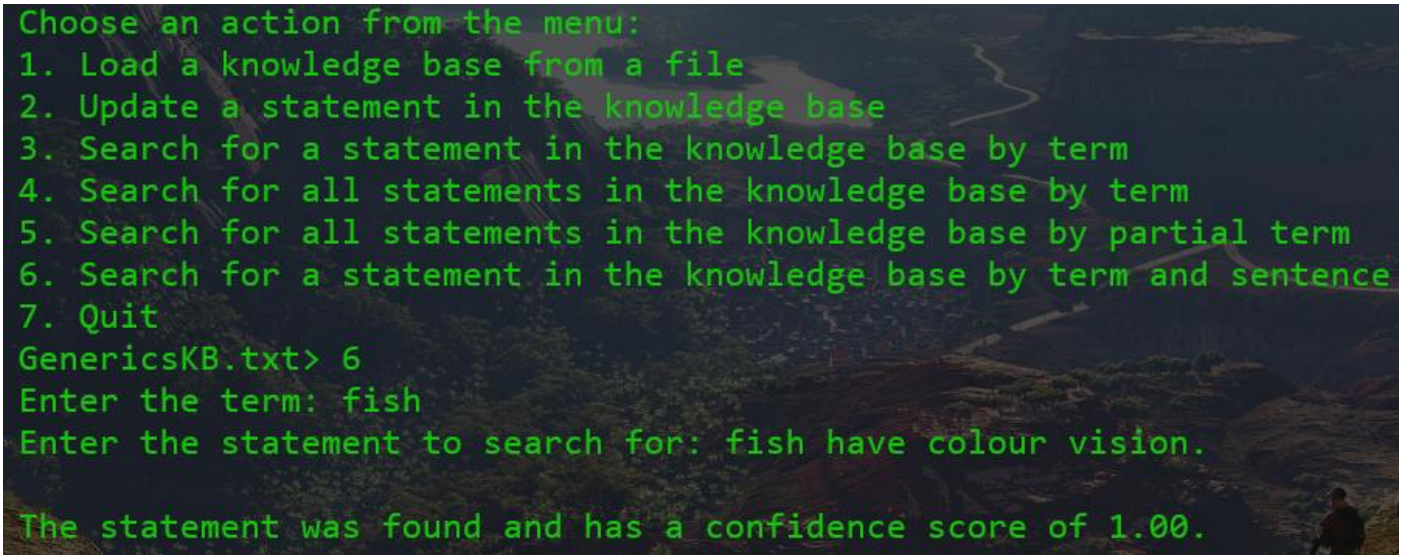
Search for all statements matching a term in the knowledge base



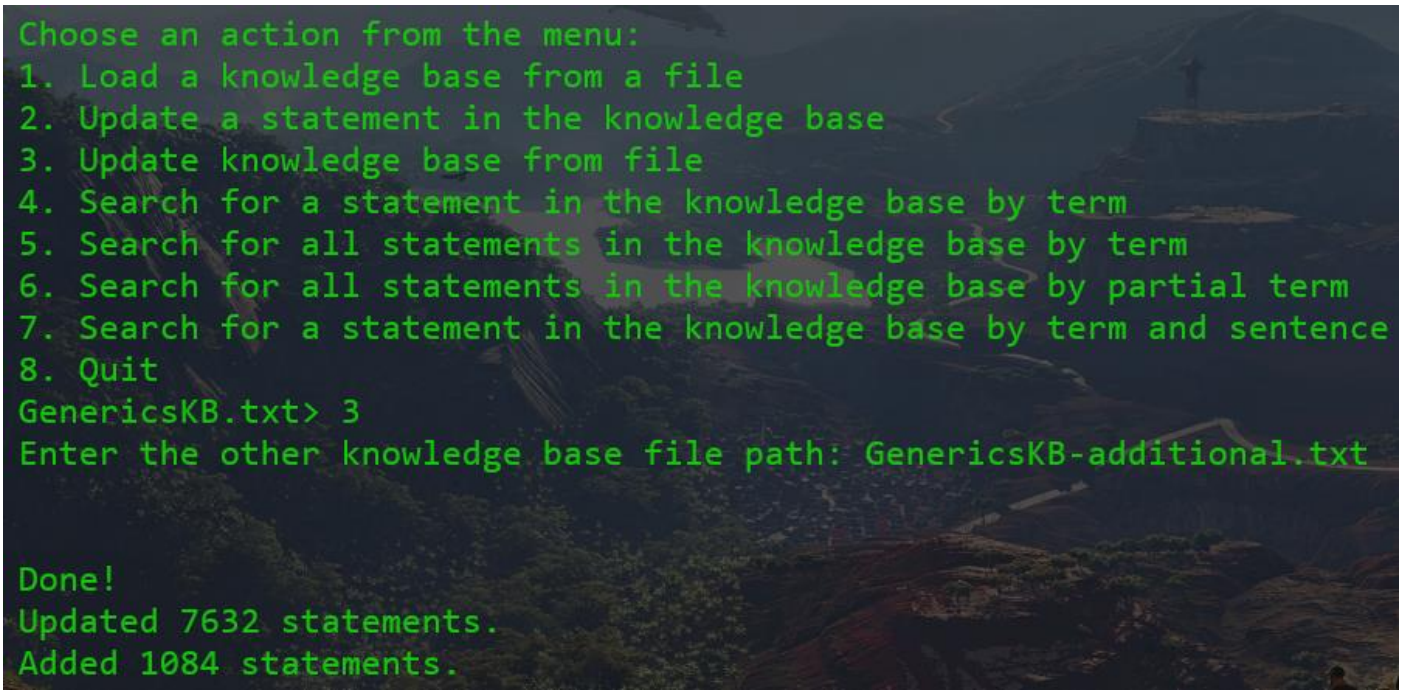
Searching for all statements in the knowledge base matching a partial term. This matches all terms beginning with the given partial term.

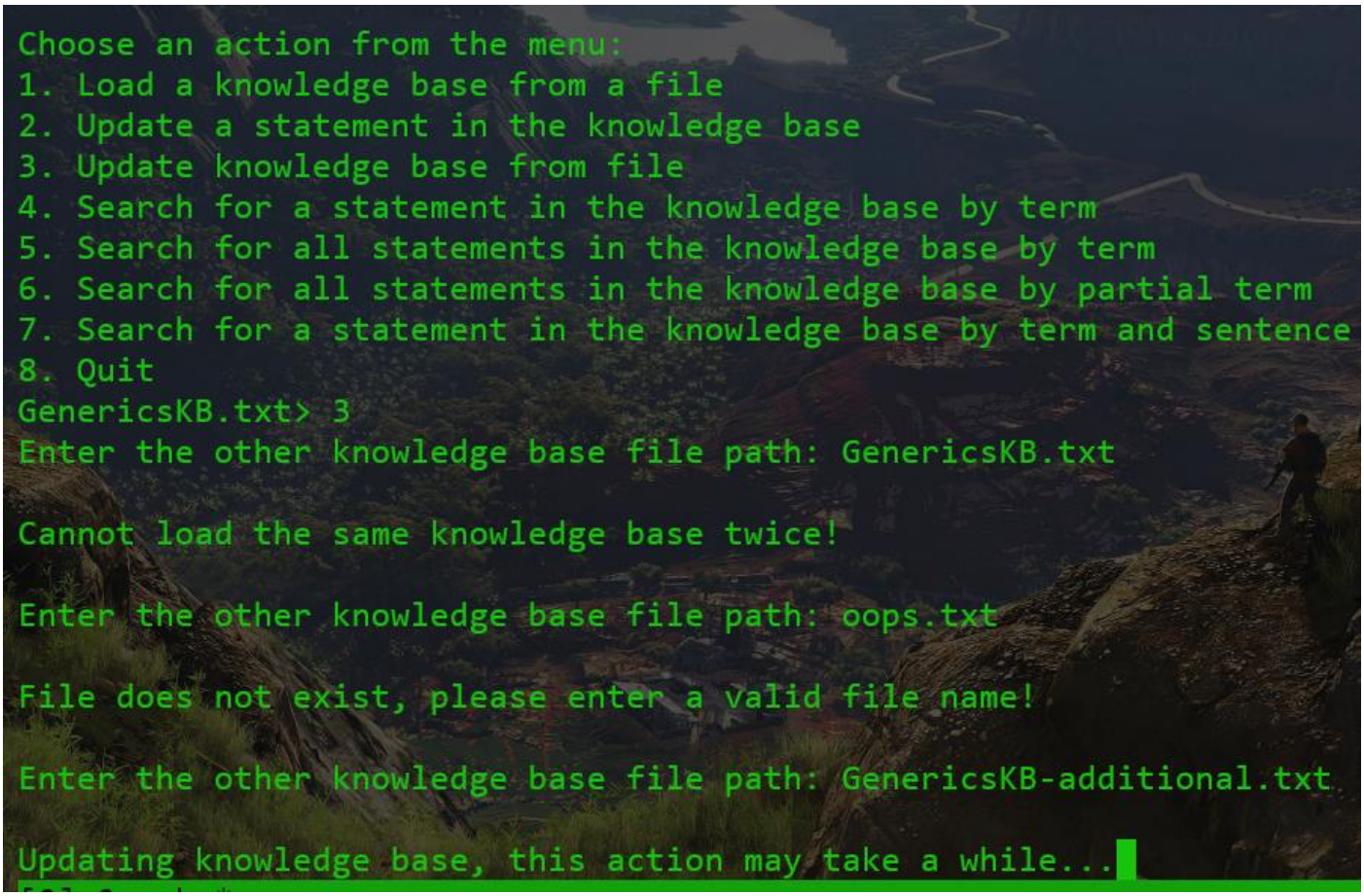


Searching the knowledge base by term and sentence.



Updating the loaded knowledge base from another knowledge base textfile. This operation does not permit the same knowledge base to be loaded twice.





Creativity

- Enhanced search functionality (option 5) to return all statements matching a specific term, as opposed to option 4 which only returns the one with the highest confidence score.
- Enhanced search functionality (option 6) to search by partial term, this search returns all statements wherein the first part of the term matches the partial term given.
- Updating the knowledge base from a file returns the number of statements updated and added to the knowledge base loaded in memory.
- The codebase exhibits modularity having a singular common behaviour between both applications. This allows changes to be made to both bst and array implementations of the app by only modifying one source file.

Git log

0: commit a4f011bc94bb9ff3a96e826dfe8301717f8073d6

1: Author: Adam <87928802+DnA-IntRicate@users.noreply.github.com>

2: Date: Mon Mar 17 12:29:20 2025 +0200

3:

4: Update KB from textfile

5:

6: commit 4f2b03f05f46e1f7fe0b24866b336592c42ef473

7: Author: Adam <87928802+DnA-IntRicate@users.noreply.github.com>

8: Date: Mon Mar 17 09:31:02 2025 +0200

9:

...

175: Author: Adam <87928802+DnA-IntRicate@users.noreply.github.com>

176: Date: Mon Mar 3 11:19:14 2025 +0200

177:

178: Create Assignments dir

179:

180: commit 49760da2adf21553f07f76764dd0b9dfe40dc75b

181: Author: Adam <87928802+DnA-IntRicate@users.noreply.github.com>

182: Date: Mon Mar 3 11:18:48 2025 +0200

183:

184: Update .editorconfig to correctly support Makefiles